

Resolved RTHP Context Decision Questionnaire

English canonical decision record for the RTHP Cross-Symbol Cycle Divergence Context. The original questionnaire was completed by explicit owner answers plus accepted recommended defaults. This document is supporting evidence; canonical authority belongs to the package manifest and versioned contracts.

Context ID: CTX_RTHP_CROSS_SYMBOL_CYCLE_DIVERGENCE_V1

Documentation release: 1.0.1

Language: English

Scope: Context only

Entry / Treatment / Execution: Out of scope

Resolved questions: 98 of 98

Unresolved domain ambiguity: 0

1. Context Boundary, Identity, and Governance

1. Is the scope of this package strictly Context, with no downstream layer defined inside it?

Accepted answer: Yes. The package is Context-only. Entry, Treatment, Execution, Risk, Position Sizing, and Trade Management are out of scope.

2. What is the current status of the Context Package?

Accepted answer: Accepted for canonical semantic reference after ACL-02 validation.

3. What is the official Context identifier?

Accepted answer: `CTX\_RTHP\_CROSS\_SYMBOL\_CYCLE\_DIVERGENCE\_V1`.

4. How is the Context versioned?

Accepted answer: Semantic Versioning. This English documentation normalization is release `1.0.1`; cycle and registry versions remain independently versioned.

5. What is the initial source of truth?

Accepted answer: The canonical manifest and versioned contracts are authoritative. The raw source and this resolved questionnaire are non-canonical evidence.

6. Who owns and independently reviews the Context?

Accepted answer: Domain owner: Amir. Independent reviewer role: RTHP Context Reviewer. Reviewer assignment may remain unassigned until governance requires a named reviewer.

2. Glossary and Fundamental Objects

7. How are Hunt, Sweep, and Break standardized?

Accepted answer: Hunt and Sweep mean the first touch of a corresponding level. Break is not a separate canonical Context concept unless introduced by a future versioned amendment.

8. How is a High-side or Low-side divergence recorded?

Accepted answer: As an independent Context event with intrinsic polarity: Low-side divergence is Bullish; High-side divergence is Bearish. It is not a trade instruction and does not guarantee reversal.

9. What is a corresponding level between the two symbols?

Accepted answer: The High or Low of the same versioned cycle definition for each symbol. Each symbol has its own numeric price.

10. Is the symbol relationship symmetric?

Accepted answer: Yes. Either symbol may become the first toucher (hunter) or the protected symbol for a particular occurrence.

11. Are fixed primary and secondary trading roles defined?

Accepted answer: No fixed semantic role exists. A pair identity is registered, and momentary hunter/protected roles are event attributes.

12. What are the canonical momentary role names?

Accepted answer: `HUNTER\_SYMBOL` and `PROTECTED\_SYMBOL`.

3. Symbol and Price-Stream Contract

13. How is the price stream used for High, Low, and touch determined?

Accepted answer: The price basis is explicit in the pair configuration. The default basis is BID.

14. What happens if the two symbols use different price bases?

Accepted answer: The relationship evaluation is invalid. A confirmed Context event cannot be created under a price-basis mismatch.

15. What exactly constitutes a High touch?

Accepted answer: The tick-size-normalized observed price is greater than or equal to the symbol-local reference High.

16. What exactly constitutes a Low touch?

Accepted answer: The tick-size-normalized observed price is less than or equal to the symbol-local reference Low.

17. What touch tolerance is used?

Accepted answer: Zero additional ticks after normalization to ``SYMBOL_TRADE_TICK_SIZE``.

18. How are concrete symbol names and broker aliases represented?

Accepted answer: The core package is generic. Each experiment registers canonical instrument IDs, broker aliases, prefixes/suffixes, contract type, and pair identity in a separate configuration artifact.

19. How is an expiring instrument identified?

Accepted answer: Every expiry is a distinct instrument identity. Continuous series, when used, must be registered separately and cannot be stitched implicitly.

4. Time, Calendar, and Cycle Boundaries

20. What is the canonical timezone?

Accepted answer: ``America/New_York``, including DST transitions.

21. How are all time intervals represented?

Accepted answer: As half-open intervals: start inclusive, end exclusive.

22. Must every timestamp belong to exactly one applicable cycle?

Accepted answer: Yes. Overlap is forbidden within the same cycle partition.

23. How is the conflict between L+N+A and the 24-hour Daily cycle resolved?

Accepted answer: They are separate concepts: ``DAYTIME_COMPOSITE`` covers L+N+A, while ``DAILY`` covers the full 20:00-to-20:00 trading day.

24. How is 20:00-07:00 represented?

Accepted answer: As a distinct overnight observation cycle inside the full Daily cycle.

25. How is Sunday 18:00-20:00 represented?

Accepted answer: As a separate partial opening segment with ``cycle_completeness=PARTIAL``.

26. How is Friday 16:00-market close represented?

Accepted answer: As a partial A segment. It is not treated as a complete A cycle by default.

27. How are holidays and early closes classified?

Accepted answer: ``COMPLETE``, ``PARTIAL``, or ``MISSING`` according to the registered New York trading calendar.

28. Which date defines the trading-day identity?

Accepted answer: The New York date on which the 20:00 Daily cycle begins.

29. Which date defines the weekly identity?

Accepted answer: The New York date on which the weekly cycle begins.

30. How is the repeated hour during the fall DST transition handled?

Accepted answer: Store absolute UTC time and the New York offset so the repeated local-hour instances remain distinct.

31. How is the missing hour during the spring DST transition handled?

Accepted answer: Use the real New York timeline. Do not create a synthetic hour.

5. Cycle Registry

32. What is the final L definition?

Accepted answer: Human presentation: 07:00:00-09:29:59 inclusive. Canonical storage: `[07:00:00, 09:30:00)`.

33. What is the final N definition?

Accepted answer: Human presentation: 09:30:00-15:59:59 inclusive. Canonical storage: `[09:30:00, 16:00:00)`.

34. What is the final A definition?

Accepted answer: Canonical storage: `[16:00:00, 20:00:00)`. Friday may close early and is then `PARTIAL`.

35. What is the final complete Daily definition?

Accepted answer: Human presentation: 20:00:00-19:59:59 inclusive. Canonical storage: `[20:00:00, 20:00:00 next day)`.

36. What is the final Weekly definition?

Accepted answer: Canonical storage: `[Sunday 18:00, Friday 17:00)` in `America/New\_York`, subject to the registered calendar and partial segments.

37. How are the two FCR subcycles defined?

Accepted answer: Human presentation: 09:30:00-09:59:59 and 10:00:00-15:59:59. Canonical storage: `[09:30,10:00)` and `[10:00,16:00)`.

38. How is the first PP subcycle defined?

Accepted answer: Human presentation: 09:00:00-09:29:59. Canonical storage: `[09:00,09:30)`.

39. Is PMI limited to L and N of the same day?

Accepted answer: Yes. PMI compares same-day L and N and has no historical lookback.

40. Is PP limited to 09:00-09:30 versus N of the same day?

Accepted answer: Yes. PP is a same-day family with no historical lookback.

41. How is a cycle labeled while still forming?

Accepted answer: `DEVELOPING` until its scheduled end, then `COMPLETE`, `PARTIAL`, or `MISSING`.

6. Reference Level, Touch, and Confirmation Time

42. From what data are cycle High and Low created?

Accepted answer: From all valid observations inside the cycle using the registered price basis.

43. How is first-touch time recorded?

Accepted answer: At tick level as immutable ``first_touch_time``, together with observation and known-time provenance.

44. What is canonical when a symbol touches the same level multiple times?

Accepted answer: The first touch is canonical. Later touches are secondary evidence or counters and do not create duplicate touch events.

45. When is the cross-symbol relationship evaluated?

Accepted answer: Touch facts are collected at tick level; the exclusive-or relationship is evaluated at each synchronized M15 close.

46. What happens if both symbols touch their corresponding level before the same M15 close?

Accepted answer: Record bilateral touch evidence and do not create an active divergence event.

47. What happens if the protected symbol touches after the confirming M15 close?

Accepted answer: The immutable historical divergence remains; its current projection becomes resolved, and the scoped reference is exhausted.

48. How are simultaneous touch timestamps handled?

Accepted answer: Record ``SIMULTANEOUS``; do not create an active divergence.

49. What happens if both High-side and Low-side relationships qualify at the same M15 cut?

Accepted answer: Create two independent Context events.

50. What happens when one symbol lacks synchronized M15 data?

Accepted answer: Retain the last observation only as ``STALE_OR_IMPUTED``; keep the relationship ``UNCONFIRMED``; set ``confirmed_context_event=false``; re-evaluate when synchronized real data arrives.

7. Context Relationship Families

51. Are WW, DD, NN, LN, AN, NL, NA, FCR, PMI, PP, and M15 Cycle Group all Context relationship-event types?

Accepted answer: Yes. All eleven are Context event families.

52. What is the official family registry?

Accepted answer: `WW`, `DD`, `NN`, `LN`, `AN`, `NL`, `NA`, `FCR`, `PMI`, `PP`, and `M15\_CYCLE\_GROUP`.

53. How many reference weeks does WW use?

Accepted answer: One prior complete eligible week by default.

54. What is the DD lookback unit?

Accepted answer: Twelve prior available complete Daily cycles.

55. What is the NN lookback unit?

Accepted answer: Twelve prior available complete N cycles.

56. What is the LN reference set?

Accepted answer: Same-day L plus up to twelve prior complete L cycles.

57. What is the AN reference set?

Accepted answer: Up to twelve prior complete A cycles; same-day A is excluded.

58. What is the NL reference set?

Accepted answer: Up to twelve prior complete N cycles; same-day N is excluded.

59. What is the NA reference set?

Accepted answer: Up to twelve complete N cycles from prior trading days; same-day N is excluded.

60. What is the FCR comparison scope?

Accepted answer: Only the two registered same-day FCR subcycles.

61. How is PMI overlap with LN handled?

Accepted answer: They remain independent families with independent event identities.

62. How is PP overlap with L and N handled?

Accepted answer: PP remains an independent family with independent event identity.

63. What happens when one M15 cut confirms multiple families?

Accepted answer: Create one independent event per qualifying family.

8. Fifteen-Minute Cycle Group

64. What is the M15 Cycle Group time domain?

Accepted answer: Only N: `[09:30,16:00)` New York time.

65. Are M15 cycle comparisons restricted to the same trading day?

Accepted answer: Yes. No cross-day M15 pair is created in version 1.

66. What exactly is sequential?

Accepted answer: Two complete M15 cycles whose indices are adjacent.

67. What exactly is non-sequential?

Accepted answer: Any complete same-N pair whose active index is more than one after the reference index.

68. What is the maximum non-sequential distance?

Accepted answer: No additional limit inside the same N session.

69. How is an incomplete terminal M15 bar handled?

Accepted answer: It remains developing and is excluded until complete.

9. Level and Occurrence State Machine

70. What is the base state of a corresponding level pair?

Accepted answer: `UNTOUCHED\_BOTH`, `ONE\_SIDE\_TOUCHED`, and `BOTH\_SIDES\_TOUCHED`, with occurrence projections such as `DIVERGENCE\_CONFIRMED`, `RESOLVED`, and `REFERENCE\_EXHAUSTED`.

71. At what scope is touch state maintained?

Accepted answer: The touch fact is global to the corresponding level pair and symbol pair. Relationship-family eligibility and occurrence identity are separately scoped.

72. What happens after BOTH_SIDES_TOUCHED?

Accepted answer: The scoped corresponding level pair cannot create a new divergence occurrence.

73. May the same High or Low support multiple families?

Accepted answer: Yes. The touch fact may be shared, while each family has independent event identity and eligibility.

74. How is a resolved divergence retained?

Accepted answer: As an immutable historical event; only its current-state projection changes.

75. When does a reference leave the candidate set?

Accepted answer: After both symbols touch the corresponding pair or when the reference leaves the configured lookback.

76. What effect do later touches by the same symbol have after first touch?

Accepted answer: They update secondary evidence or counters only; they do not create a new touch event.

77. What if the same numeric price appears in a different reference cycle?

Accepted answer: A new reference-cycle identity creates a new level and state. Numeric equality does not merge identities.

10. Partial History, Synchronization, and Reconstruction

78. What unit counts lookback?

Accepted answer: Available eligible trading cycles, not calendar days or a fixed bar count.

79. What happens when available history is shorter than requested lookback?

Accepted answer: Record `PARTIAL\_DATA` and evaluate every available eligible reference.

80. What happens when no reference cycle exists?

Accepted answer: Record `INSUFFICIENT\_DATA` and create no relationship event.

81. How is a Saturday absent from the chart handled?

Accepted answer: Do not create a synthetic Saturday cycle.

82. How is a missing bar in one symbol handled?

Accepted answer: The dependent cycle or evaluation becomes `BLOCKED\_BY\_DATA`; no forward-fill or synthetic bar is allowed.

83. How are late or out-of-order ticks handled?

Accepted answer: Store source time and known time separately. Late evidence creates an append-only revision with provenance.

84. How is rollover handled for expiring contracts?

Accepted answer: Each contract keeps independent state; implicit stitching and state transfer are forbidden.

85. What happens when MetaTrader history is corrected or reloaded?

Accepted answer: Append a revision event with provenance. Do not delete or silently overwrite the prior event.

11. Identity, Deduplication, and Parity

86. What components form a touch-event ID?

Accepted answer: Context ID, pair ID, symbol ID, reference-cycle ID, level side, first-touch time, and data-source identity.

87. What components form a relationship-event ID?

Accepted answer: Context ID, pair ID, cycle-definition version, family, active-cycle ID, reference-cycle ID, level side, hunter/protected roles, first-touch time, and confirmation-close time.

88. What is the post-restart deduplication policy?

Accepted answer: The same event ID is never appended twice.

89. How is state reconstructed after restart?

Accepted answer: From the immutable event ledger and deterministic projections, never from chart drawings or unproven cache state.

90. What historical/live-like parity is required?

Accepted answer: Event identity, first touch, confirmation cut, state transition, data sufficiency, and exhaustion must match.

91. What happens when historical and live-like paths disagree?

Accepted answer: Create a mismatch record and block Context acceptance until resolved.

12. Context Output Boundary and Required Evidence

92. What canonical artifacts must the package emit?

Accepted answer: Cycle registry, reference-level contract, touch ledger contract, pair-state model, relationship events, data-sufficiency states, UNKNOWN/blocker policy, provenance, schemas, and acceptance fixtures.

93. Do all presentation rules remain outside the Context Package?

Accepted answer: Yes. Context emits facts and state only. Historical event persistence belongs to the ledger; visual style belongs to a downstream visualizer.

94. Does downstream meaning remain outside the package?

Accepted answer: Trading and predictive meaning remain outside. The package may record intrinsic Bullish/Bearish Context polarity, but it does not authorize Entry, guarantee reversal, or define a trade side.

95. May UNKNOWN values be converted automatically to defaults?

Accepted answer: No. UNKNOWN remains explicit and blocking where required.

96. Which scenarios require exact numeric fixtures?

Accepted answer: Bilateral touch in one M15, later protected touch, multiple families in one cut, partial Friday, partial Sunday, DST transitions, zero-history expiry, sequential M15, and non-sequential M15.

97. What repository path, commit, and file authoritatively implement the prior reference-consumption rule?

Accepted answer: None was provided. Prior code is non-authoritative evidence only.

98. Was any additional Context rule left uncovered?

Accepted answer: No unresolved domain ambiguity remains after the accepted decisions and canonical contract review.

End of resolved questionnaire. No rejected option is authoritative. Explicit owner decisions override recommended defaults; all other unanswered items adopt the accepted recommendation.